Syllabus Details

Syllabus ID:	13144
Syllabus Name:	**Toán rời rạc**
Course Name English:	**Discrete mathematics**
Subject Code:	**MAD101**
Learning-Teaching Method:	Lecture,Project based learning,Problem based learning,Social Constructivism
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45 contact hours (60 sessions) + 1 hour final exam + 104 hours self-study
Pre-Requisite:	None
Description:	Upon finishing the course, students must acquire: 1. the following knowledge: (ABET a.1) • Concepts of logical expressions & predicate logic. • The method of induction and recursive definition. • Concepts of algorithms, recursive algorithms, the complexity. • Recurrence relations and divide-and-conquer algorithms. • Application of integers and congruence in information technology. • Set structure and map, counting principles and combinatorics concepts. • The terminologies and properties of graphs & trees & weighted graphs. • The applications of graphs, trees and weighted graphs in information technology. 2. the following skills: (ABET a.2) • Manipulate logical expressions & produce equivalent logical expressions. • Produce proof of simple mathematical propositions, including applying induction method. • Evaluate complexity of algorithms. • Solving simple congruence equations. • Apply counting techniques in evaluating various set cardinalities. • Solve counting problems, apply in analyze algorithms. • Apply graph's algorithms to solve problems related to graph theory.
StudentTasks:	Students must attend at least 80% of contact hours in order to be accepted to the final examination.
Tools:	- Testbank for ST1. - Powerpoint image in CD + Powerpoint prepared by previous lecturers (PP). - Computerized tools (for example C programming tools, Maxima http://maxima.sourceforge.net/) (CT) - Teaching Suggestion and Applications of Discrete Mathematics for ST1 (TS).
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

6 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Adam Drozdek. Data structure and Algorithms in Java. 3rd Edition.						False	True	False	
2	Computerized tools (for example C programming tools, Maxima http://maxima.sourceforge.net/) (CT)						False	False	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Discrete Mathematics and its applications	Kenneth Rosen	Mc.Graw Hill, IE	2007	Ed.7 (or 6)	978-0073383095	True	True	False	
4	Powerpoint image in CD + Powerpoint prepared by previous lecturers (PP)						False	False	False	
5	Teaching Suggestion and Applications of Discrete Mathematics for ST1 (TS)						False	False	False	
6	Testbank for ST1						False	False	False	

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Manipulate logical expressions and use logic for inference LO1.1 Determine the truth value of a proposition; check if two propositions are equivalent; translate between logical expressions and sentences for propositional logic. LO1.2 Determine the truth value of a proposition; check if two propositions are equivalent; translate between logical expressions and sentences for predicate logic. LO1.3 Manipulate logical expressions with nested quantifiers. LO1.4 Recognize the inference rules used in a valid statemment; use inference rules to draw conclusions; check if a statement is valid.
2	CLO2	Perform basic operations on sets, functions and sequences LO2.1 Distinguish basic terminologies in set theory; find the power set and cartersian product; represent a subset using binary bits. LO2.2 Perform operations on sets; check if two sets are equal. LO2.3 Determine if a given rule represents a function; perform operations on functions and manipulate functions inlvolving the floor and the ceiling functions. LO2.4 Determine if a function is one-to-one, onto or a bijection, and describe how these properties can be used to compare two sets. LO2.5 Find a formula for a sequence; evaluate a finite sum with a possible use of special summations.
3	CLO3	Analyze basic algorithms written in pseudo codes and determine their complexities LO3.1 Determine if a procedure is an algorithm; work out basic algorithms written in pseudocodes including searching, sorting and greedy algorithms. LO3.2 Find big-O, big-Omega and big-Theta estimates of a function; describe complexity of an algorithm.
4	CLO4	Discuss applications of modular arithmetic and analyze algorithms on integers LO4.1 Perform integer division and modular arithmetic; apply congruence in generating pseudorandom numbers and in cryptology. LO4.2 Find the base b expansion of an integer; perform binary operations and modular exponentiation on integers. LO4.3 Check if a number is prime and find prime factorization of an integer; find greatest common divisor and least common multiple.
5	CLO5	Use induction and recursive definition to prove a statement or to describe an object, and write recursive algorithms LO5.1 Prove mathematical statements by induction. LO5.2 Manipulate sequences and sets defined by recursion. LO5.3 Work out recursive algorithms, including the mergesort algorithm; distinguish between iterative and recursive algorithms.
6	CLO6	Solve counting problems LO6.1 Use basic rules for counting LO6.2 Use recurrence relations for counting, including divide-and-conquer recurrence relations.
7	CLO7	Analyze the structure of a graph and find shortest paths in a weighted graph LO7.1 Classify and describe a graph; use hand-shaking theorem to analyze a graph; identify special graphs including bipartite graphs. LO7.2 Represent a graph using incidence and adjacency matrices; determine if two graphs are isomorphic. LO7.3 Check connectivity of a graph and identify its connected components; count paths in a graph. LO7.4 Determine if a graph has Euler/Hamilton paths/circuits; find Euler and Hamilton paths/circuits in a graph. LO7.5 Run Dijsktra's algorithm to determine the shortest paths and the shortest distance in a weighted graph.
8	CLO8	Analyze the structure of a tree and describe algorithms using trees LO8.1 Check if a graph is a tree, and identify basic terminologies of a tree; use properties of m-ary trees to solve problems. LO8.2 Build binary search trees and decision trees; analyze complexities of algorithms described using binary search trees and decision trees. LO8.3 Recognize and construct prefix codes; build optimal trees using Huffman coding. LO8.4 Traverse a tree and use tree traversal to generate notations; evaluate prefix and postfix expressions. LO8.5 Run Depth-first search and Breadth-first search algorithms to find spanning trees LO8.6 Find minimum spanning trees using Prim, Kruskal,... algorithms

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	0 Introduction 1.1 Propositional Logic (1.1 and 1.2, skip the topic on Logic Circuits)	Offline	LO1	TU	Textbook Chapter 1	MAD101_Part1	Exercises Chapter 1	
2	1.2 Propositional Equivalences (1.3, skip the topic on Satisfiability)	Offline	LO1	TU	Textbook Chapter 1	MAD101_Part2	Exercises Chapter 1	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	1.3 Predicates (1.4)	Offline	LO1	TU	Textbook Chapter 1		Exercises Chapter 1	
4	1.3 Quantifiers	Offline	LO1	TU	Textbook Chapter 1		Exercises Chapter 1	
5	1.4 Nested Quantifiers (1.5)	Offline	LO1	TU	Textbook Chapter 1		Exercises Chapter 1	
6	1.4 Nested Quantifiers (1.5) (cont.)	Offline	LO1	TU	Textbook Chapter 1		Exercises Chapter 1	
7	1.5 Rules of Inference (1.6)	Offline	LO1	TU	Textbook Chapter 1		Exercises Chapter 1	
8	1.5 Rules of Inference (1.6) (cont.)	Offline	LO1	IT	Textbook Chapter 1		Exercises Chapter 1	
9	2.1 Introduction to sets	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2	
10	2.2 Set operations	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2	
11	2.3 Functions (2.3, skip the topic on Partial Functions)	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2	
12	2.3 Functions (2.3, skip the topic on Partial Functions) (cont.)	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2	
13	2.4 Sequences and Summations (2.4, skip the topic on Recurrence relations)	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2	
14	2.4 Sequences and Summations (2.4, skip the topic on Recurrence relations) (cont.)	Offline	LO2	TU	Textbook Chapter 2		Exercises Chapter 2	
15	5.1 Basic Counting Principles (6.1)	Offline	LO6	TU	Textbook Chapter 5		Exercises Chapter 5	
16	5.1 Basic Counting Principles (6.1) (continued)	Offline	LO6	TU	Textbook Chapter 5		Exercises Chapter 5	
17	Review	Offline	LO1, LO2, LO6					
18	Progress Test 1	Offline	LO1, LO2, LO6					
19	3.1 Algorithms	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3	
20	3.2 The Growth of Functions	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3	
21	3.3 Complexity of Algorithms (3.3, skip the topics on Matrix multiplication, Algorithmic Paradigms)	Offline	LO3	TU	Textbook Chapter 3		Exercises Chapter 3	
22	3.4 The Integers and Division (4.1, 4.5, and the topic on Classical Cryptography 4.6)	Offline	LO4	TU	Textbook Chapter 3		Exercises Chapter 3	
23	3.4 The Integers and Division (4.1, 4.5, and the topic on Classical Cryptography 4.6) (cont.)	Offline	LO4	TU	Textbook Chapter 3		Exercises Chapter 3	
24	3.5 Primes and Greatest Common Divisors (4.3)	Offline	LO4	TU	Textbook Chapter 3		Exercises Chapter 3	
25	3.5 Primes and Greatest Common Divisors (4.3) (cont.)	Offline	LO4	TU	Textbook Chapter 3		Exercises Chapter 3	
26	3.6 Intergers and Algorithms (4.2)	Offline	LO4	TU	Textbook Chapter 3		Exercises Chapter 3	
27	3.6 Intergers and Algorithms (4.2) (cont.)	Offline	LO4	TU	Textbook Chapter 3		Exercises Chapter 3	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
28	3.6 Intergers and Algorithms (4.2) (cont.)	Offline	LO4	TU	Textbook Chapter 3		Exercises Chapter 3	
29	4.1 Mathematical Induction (5.1)	Offline	LO5	IT	Textbook Chapter 4		Exercises Chapter 4	
30	4.2 Strong Induction (5.2)	Offline	LO5	IT	Textbook Chapter 4		Exercises Chapter 4	
31	4.3 Recursive Definitions and Structural Induction (5.3)	Offline	LO5	IT	Textbook Chapter 4		Exercises Chapter 4	
32	4.3 Recursive Definitions and Structural Induction (5.3) (cont.)	Offline	LO5	IT	Textbook Chapter 4		Exercises Chapter 4	
33	4.4 Recursive Algorithms (5.4)	Offline	LO5	TU	Textbook Chapter 4		Exercises Chapter 4	
34	4.4 Recursive Algorithms (5.4) (cont.)	Offline	LO5	TU	Textbook Chapter 4		Exercises Chapter 4	
35	7.1 Recurrence Relations (8.1, skip the topic on Dynamic programming)	Offline	LO6	TU	Textbook Chapter 7		Exercises Chapter 7	
36	7.3 Divide-and-Conquer Algorithms and Recurrence Relations (8.3)	Offline	LO6	IT	Textbook Chapter 7		Exercises Chapter 7	
37	Review	Offline		U				
38	Progress Test 2	Offline	LO3, LO4, LO5, LO6	U				
39	9.1 Graphs and Graph Models (10.1)	Offline	LO7	I	Textbook Chapter 9		Exercises Chapter 9	
40	9.2 Graph Terminology and Special Types of Graphs	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
41	9.3 Representing Graphs and Graph Isomorphism (10.3)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
42	9.3 Representing Graphs and Graph Isomorphism (10.3) (cont.)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
43	9.4 Connectivity (10.4)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
44	9.4 Connectivity (10.4) (cont.)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
45	9.5 Euler and Hamilton Paths (10.5)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
46	9.5 Euler and Hamilton Paths (10.5) (cont.)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
47	9.6 Shortest-Path Problems (10.6)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
48	9.6 Shortest-Path Problems (10.6) (cont.)	Offline	LO7	TU	Textbook Chapter 9		Exercises Chapter 9	
49	10.1 Introduction to Trees (11.1)	Offline	LO8	IT	Textbook Chapter 10		Exercises Chapter 10	
50	10.1 Introduction to Trees (11.1) (cont.)	Offline	LO8	IT	Textbook Chapter 10		Exercises Chapter 10	
51	10.2 Applications of Trees (11.2)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	
52	10.2 Applications of Trees (11.2) (cont.)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	
53	10.3 Tree Traversal (11.3)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
54	10.3 Tree Traversal (11.3) (cont.)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	
55	10.4 Spanning Trees (11.4)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	
56	10.4 Spanning Trees (11.4) (cont.)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	
57	10.5 Minimum Spanning Trees (11.5)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	
58	10.5 Minimum Spanning Trees (11.5) (cont.)	Offline	LO8	TU	Textbook Chapter 10		Exercises Chapter 10	
59	Review	Offline	LO7, LO8	U				
60	Progress Test 3	Offline	LO7, LO8	U				

166 Constructive question(s)

No.	Session No	Name	Details
1	1	CT_CQ1	What are the types of propositional logic?
2	1	CQ1_HCM	List as many logical operators as possible. Give examples that use operators you listed.
3	2	CT_CQ1	What does it mean for two propositions to be logically equivalent? Describe the different ways to show that two compound propositions are logically equivalent.
4	2	CQ1_HCM	How do we define the concept 'logical equivalence' in propositional logic? Let's give examples.
5	2	CQ2_HCM	Give two examples using De Morgan laws.
6	3	CT_CQ1	How to understand the universal and existential quantifications of a predicate P(x)? What are their negations?
7	3	CQ1_HCM	What is the main different between a predicate and a proposition? Illustrate your statement by examples.
8	4	CQ1_DN	What are two most important quantifiers? How is Demorgan's law in predicate logic understood?
9	4	CT_CQ1	What is the precedence of quantifiers?
10	4	CQ1_HCM	In what case is the value of the proposition $\forall x P(x)$ false? Give counterexamples.
11	5	CQ1_DN	What are two most important quantifiers? How is Demorgan's law in predicate logic understood?
12	5	CT_CQ1	What is the difference between the quantification $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$, where P(x, y) is a predicate? Give an example of a predicate P(x, y) such that they have different truth values?
13	5	CQ1_HCM	Let P(x, y) be a predicate. Give an instance that makes the truth value of the propositions $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$ true/false.
14	6	CQ1_DN	What are two most important quantifiers? How is Demorgan's law in predicate logic understood?
15	6	CT_CQ1	How to negate a statement involving nested quantifiers? Give examples.
16	6	CQ1_HCM	What is the negation of $\exists x \forall y(P(x, y) \rightarrow Q(x,y))$? Give examples in English statements.
17	6	CQ2_HCM	Give examples of translation between English statements and logical expressions.
18	7	CT_CQ1	Describe what is meant by a valid argument in propositional logic and show that the argument “If the earth is flat, then you can sail off the edge of the earth,” “You cannot sail off the edge of the earth,” therefore, “The earth is not flat” is a valid argument
19	7	CQ1_HCM	List some rules of inference. Use them to give your own valid arguments.
20	8	CT_CQ1	Use rules of inference to show that if the premises “All zebras have stripes” and “Mark is a zebra” are true, then the conclusion “Mark has stripes” is true.
21	8	CQ1_HCM	For each of these arguments, determine whether it is correct or incorrect, and explain why. Give your correction to obtain a valid one. a) Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major. b) Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.
22	9	CT_CQ1	Explain what it means for one set to be a subset of another set. How do you prove that one set is a subset of another set?

No.	Session No	Name	Details
23	9	CQ1_HCM	Explain the concepts: sets, infinite sets, subsets of a set, power set of a set. Give examples.
24	9	CQ2_HCM	List some set operators with examples. What do we mean by 'symmetric difference' in set theory?
25	10	CQ1_DN	How are two or more sets combined? What are the important properties of functions?
26	10	CT_CQ1	What are the union, intersection, difference, and symmetric difference of the set of positive integers and the set of odd integers?
27	10	CQ1_HCM	Is it possible that A is an element of A, where A is a set? Give examples.
28	10	CQ2_HCM	How do we represent sets in computers? Give examples for illustration.
29	11	CQ1_DN	How are two or more sets combined? What are the important properties of functions?
30	11	CT_CQ1	Give some examples of functions from the set of positive integers to the set of positive integers being: 1) both one-to-one and onto 2) one-to-one but not onto 3) onto but not not one-to-one 4) neither one-to-one nor onto
31	11	CQ1_HCM	What are functions? one-to-one functions? How many one-to-one functions are from a set of three elements to itself?
32	12	CQ1_DN	How are two or more sets combined? What are the important properties of functions?
33	12	CT_CQ1	When does a function have an inverse?
34	12	CQ1_HCM	When is a function called bijection and what is its inverse?
35	13	CT_CQ1	For each of the following lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence. (a) 3, 6, 11, 18, 27, 38, 51, 66, 83, 102, ... (b) 15, 8, 1, −6, −13, −20, −27, ... (c) 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, ... (d) 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, ...
36	13	CQ1_HCM	Give your analysis to know what the 50th term is. 1, 2, 3, 3, 4, 5, 5, 5, 6, 7, 7, 7, 7, ...
37	13	CQ2_HCM	What is a geometric progression? Give an example.
38	14	CT_CQ1	How to evaluate the double sum?
39	14	CQ1_HCM	What is the sum of the first 50 terms of the following sequence 1, 2, 3, 3, 4, 5, 5, 5, 6, 7, 7, 7, 7, ...?
40	14	CQ2_HCM	Can infinite sets be countable? Why or why not?
41	15	CT_CQ1	Calculate the number of one-to-one functions from a set S to a set T, where S and T are finite sets of various sizes. Can you determine a formula for the number of such functions?
42	16	CQ1_DN	How to locate an element in an ordered list? What is commonly used to describe the growth of a function?
43	17	CQ1_DN	How to locate an element in an ordered list? What is commonly used to describe the growth of a function?
44	17	CT_CQ1	What is the difference between an algorithm and a computer program?
45	17	CQ1_HCM	Give at least 3 different algorithms to compute 1+2+3+...+n for positive integer n.
46	17	CQ2_HCM	Give an algorithm that the number of basic comparisons does not depend on the size of input.
47	18	CQ1_DN	How to locate an element in an ordered list? What is commonly used to describe the growth of a function?
48	18	CT_CQ1	What is big-O notation used for?
49	18	CQ1_HCM	Give at least 3 different algorithms to compute 1+2+3+...+n for positive integer n. Compute the number of additions used in each algorithm.
50	19	CT_CQ1	How can the efficiency of an algorithm be analyzed?
51	19	CQ2_HCM	Show that log(n!) is O(nlogn).
52	20	CT_CQ1	Let n and d be positive integers. How many positive integers not exceeding n are divisible by d?
53	20	CQ1_HCM	Is it possible to use only paper and pencil to calculate 3^2023 mod 78? Give your solution if any.
54	21	CT_CQ1	Define what it means for a and b to be congruent modulo m?
55	21	CQ1_HCM	Explain how to encrypt the message 'HAPPY' using modular arithmetic with the encryption function f(p) = (p + 7) mod 26.
56	21	CQ2_HCM	Is the function f(p) = (13p + 7) mod 26 good for encoding? Why or why not?
57	22	CQ1_DN	How many ways are there to find the greatest common divisor of two numbers? What are the algorithms for adding and mutiplying integers using their base b expansions?
58	22	CT_CQ1	Describe a procedure for finding the prime factorization of an integer

No.	Session No	Name	Details
59	22	CQ1_HCM	Why are primes important in cryptography?
60	23	CQ1_DN	How many ways are there to find the greatest common divisor of two numbers? What are the algorithms for adding and mutiplying integers using their base b expansions?
61	23	CT_CQ1	Describe some ways to find the greatest common divisor of two integers. When does each method work best?
62	23	CQ1_HCM	Give examples to illustrate that if r = a mod b then gcd(a, b) = gcd(b, r).
63	23	CQ2_HCM	Explain the Euclidean algorithm via examples.
64	24	CQ1_DN	How many ways are there to find the greatest common divisor of two numbers? What are the algorithms for adding and mutiplying integers using their base b expansions?
65	24	CT_CQ1	Describe a procedure for converting decimal (base 10) expansions of integers into a base n (n>1) expansions and vice versa.
66	24	CQ1_HCM	Give an example to explain how to find a binary format of a positive integer.
67	24	CQ2_HCM	Can an integer have base 3 expansion? Give an example.
68	25	CT_CQ1	Describe an algorithm for finding the difference of two binary expansions?
69	25	CQ1_HCM	Explain ALGORITHM 3 Multiplication of Integers. (page 266, K. Rosen 7th Ed.)
70	26	CT_CQ1	How can we find 3^2022 mod 99 using only paper and pencil?
71	26	CQ1_HCM	How can we compute 7^333 mod 555 using Fast Modular Exponentiation algorithm? How fast is this algorithm?
72	27	CT_CQ1	Can you use the principle of mathematical induction to find a formula for the sum of the first n terms of a sequence?
73	27	CQ1_HCM	Give an example of a recursive algorithm.
74	27	CQ2_HCM	Use mathematical induction to prove the correctness of a recursive algorithm of computing n!
75	28	CT_CQ1	Which amounts of money can be formed using just two dollar bills and five-dollar bills?
76	28	CQ1_HCM	Use strong induction to prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.
77	29	CT_CQ1	Give a recursive definition of a^n, where a is a nonzero real number and n is a nonnegative integer?
78	29	CQ1_HCM	Give at least two recursive definitions for the sequence {a_n} with n>0: 0, 5, 0, 5, 0, 5, ...
79	30	CT_CQ1	Give a recursive definition of the set of positive integers that are multiples of 5.
80	30	CQ1_HCM	List five elements of the set of all strings generated from {a, b}.
81	30	CQ2_HCM	Use a recursive definition to define the set of all strings generated from {a, b}.
82	31	CQ1_DN	Can we solve a problem by reducing it to an instance of the same problem with smaller input? What are the basic rules of counting?
83	31	CT_CQ1	Describe a recursive algorithm for 1) computing n! 2) computing a^n 3) computing the greatest common divisor of two positive integers. 4) linear searching the value x in a subsequence of the sequence {a_n}. 5) binary searching the value x in an increasingly ordered subsequence of the sequence {a_n}.
84	31	CQ1_HCM	Write recursive versions of linear search algorithm.
85	31	CQ2_HCM	Write recursive versions of binary search algorithm.
86	32	CQ1_DN	Can we solve a problem by reducing it to an instance of the same problem with smaller input? What are the basic rules of counting?
87	32	CT_CQ1	Estimate the number of comparisons used to merge two ordered lists in the analysis of the merge sort.
88	32	CQ1_HCM	Write an interactive algorithm and a recursive algorithm to find the Fibonacci numbers. Which algorithm uses fewer comparisons?
89	33	CQ1_DN	Can we solve a problem by reducing it to an instance of the same problem with smaller input? What are the basic rules of counting?
90	33	CT_CQ1	What is the difference between: a) The sum rule and product rule? b) The permutation and combination?
91	33	CQ1_HCM	State the principle of inclusion-exclusion. Give an example where this principle applies.
92	33	CQ2_HCM	State the principle of inclusion-exclusion for three sets.

No.	Session No	Name	Details
93	34	CQ1_DN	How recurrence relations can be used to solve counting problems? How to use these divide-and-conquer recurrence relations to estimate the complexity of these algorithms?
94	34	CT_CQ1	Explain how the principle of inclusion–exclusion can be used to count the number of onto functions from a set with m elements to a set with n elements.
95	34	CQ1_HCM	How many bit strings of length six do not have two consecutive 1s or three consecutive 0s?
96	35	CQ1_DN	How recurrence relations can be used to solve counting problems? How to use these divide-and-conquer recurrence relations to estimate the complexity of these algorithms?
97	35	CT_CQ1	Explain how the Fibonacci numbers are used to find a recurrence relation for the number of bit strings of length n not containing two consecutive 1s.
98	35	CQ1_HCM	State the product rule for counting. Give 3 examples.
99	35	CQ2_HCM	Give examples of counting problems which can be solved by using tree diagrams.
100	36	CQ1_DN	How recurrence relations can be used to solve counting problems? How to use these divide-and-conquer recurrence relations to estimate the complexity of these algorithms?
101	36	CT_CQ1	Explain how to find f(b^k) (where k is a positive integer) if f(n) satisfies the divide-and-conquer recurrence relation f(n) = af (n/b) + g(n) whenever b divides the positive integer n
102	36	CQ1_HCM	Given f(n) = 2f(n/2) + n when n is an integer divisible by 2 and f(1) = 1. Can we compute the exact value of f(1024)? Why or why not? How do we estimate the growth of f(n)?
103	37	CT_CQ1	Derive divide-and-conquer recurrence relations for the number of comparisons used in: binary search; locating the maximum element in a sequence. Give big-O estimates for the number of comparisons needed by these algorithm.
104	39	CT_CQ1	Explain how to model different aspects of a computer network or airline routes.
105	39	CQ1_HCM	List some models of graphs in the real world.
106	39	CQ2_HCM	Explain the concepts: degree, in/out-degree, adjacency and incident.
107	40	CQ1_DN	What are the number of vertices and edges of several classes of simple graphs often used in many applications? How to represent graphs?
108	40	CT_CQ1	What is the relationship between the sum of the degrees of the vertices in an undirected graph and the number of edges in this graph?
109	40	CQ1_HCM	Describe some special simple graphs Kn, Cn, Wn
110	40	CQ2_HCM	Describe some special simple graphs Qn, Km;n.
111	41	CQ1_DN	What are the number of vertices and edges of several classes of simple graphs often used in many applications? How to represent graphs?
112	41	CT_CQ1	How to represent graphs in several different ways?
113	41	CQ1_HCM	How to generate the adjacency matrix and incidence matrix from a given graph, and vice versa?
114	41	CQ2_HCM	Highlights the difference between adjacency matrix and incidence matrix.
115	42	CQ1_DN	What are the number of vertices and edges of several classes of simple graphs often used in many applications? How to represent graphs?
116	42	CT_CQ1	How to determine whether two graphs are isomorphic?
117	42	CQ1_HCM	Give an example of two non-isomorphic graphs that have the same number of nodes, edges and degree sequences.
118	43	CT_CQ1	When is there always a path between two vertices in the graph?
119	43	CQ1_HCM	What is the length of the longest simple circuit in Kn, Wn?
120	44	CT_CQ1	How can paths and circuits help to determine whether two graphs are isomorphic?
121	44	CQ1_HCM	What are cut vertices and cut edges?
122	44	CQ2_HCM	Explain the difference between strongly connected and weakly connected. Illustrate them by graphs.
123	45	CT_CQ1	Describe the famous Konigsberg bridge problem and explain how to rephrase it in terms of an Euler circuit.
124	45	CQ1_HCM	What are Euler Circuits/Paths? What is the indication that a given graph has an Euler Circuit/Path?
125	45	CQ2_HCM	Can L. Euler solve the Seven Bridges of Königsberg Problem?

No.	Session No	Name	Details
126	46	CQ1_DN	How to find a shortest path between two vertices in a weighted graph using a greedy approach? Can we solve the traveling salesman problem with a backtracking technique?
127	46	CT_CQ1	Give some properties of a simple graph that imply that it has a Hamilton circuit.
128	46	CQ1_HCM	In a weighted graph, assume that the weights of the edges are different. Is the shortest path between two vertices unique? Show your claim by using examples.
129	47	CQ1_DN	How to find a shortest path between two vertices in a weighted graph using a greedy approach? Can we solve the traveling salesman problem with a backtracking technique?
130	47	CT_CQ1	Describe Dijkstra's algorithm for finding the shortest path in a weighted graph between two vertices.
131	47	CQ1_HCM	Use Djikstra's algorithm to find the shortest path and also its length from a source vertex to a sink vertex in a weighted graph. Describe each iteration of the algorithm in detail.
132	47	CQ2_HCM	Does Google Map use Djikstra's algorithm?
133	48	CQ1_DN	How to find a shortest path between two vertices in a weighted graph using a greedy approach? Can we solve the traveling salesman problem with a backtracking technique?
134	48	CT_CQ1	Draw a weighted graph with at least 8 vertices and 12 edges. Use Dijkstra's algorithm to find the shortest path between two vertices of your choice in the graph.
135	48	CQ1_HCM	Explain why the exhausted technique cannot be used to solve the TSP (Traveling Salesman Problem) when the number of nodes is large.
136	49	CT_CQ1	Draw a rooted tree with at least 10 vertices, where the degree of each vertex does not exceed 3. Identify the root, the parent of each vertex, the children of each vertex, the internal vertices, the leaves, the level of each vetex, and the height of tree.
137	49	CQ1_HCM	Give some tree models that are commonly used in the real world.
138	49	CQ2_HCM	Explain the concepts parent, child, leaf, internal nodes, descendants, and ancestors used in graph theory.
139	50	CT_CQ1	How many edges does a tree with n vertices have? How many vertices and leaves does a full m-ary tree have if it has i internal vertices? How many leaves can an m-ary tree of height h have?
140	50	CQ1_HCM	A full binary tree has L leaves. How high can this tree be? If the number of nodes is given by n, then what are the maximum and minimum heights of a binary tree?
141	51	CT_CQ1	What is a binary search tree? Describe an algorithm for constructing a binary search tree, and form a binary search tree for the words: vireo, warbler, egret, grosbeak, nuthatch, and kingfisher.
142	51	CQ1_HCM	Build a binary search tree for the words enology, phrenology, campanology, ornithology, ichthyology, limnology, alchemy, and astrology using alphabetical order. How many comparisons are needed to locate/find or insert/add the word "limnology" in the search tree?
143	52	CQ1_DN	How to code a text using a prefix tree? What is the algorithm to convert from an expression from infix form to postfix one?
144	52	CT_CQ1	What is a prefix code? How can a prefix code be represented by a binary tree?
145	52	CQ1_HCM	Use the Huffman coding algorithm to encode a letter given the frequencies A: 0.1 B: 0.13 C: 0.22 D: 0.25 E: 0.30.
146	52	CQ2_HCM	What is the main benefit of Huffman coding algorithm?
147	53	CQ1_DN	How to code a text using a prefix tree? What is the algorithm to convert from an expression from infix form to postfix one?
148	53	CT_CQ1	Describe preorder, inorder, and postorder tree traversal. Give some examples of preorder, postorder, and inorder traversal of some binary trees of your choice with at least 10 vertices.
149	53	CQ1_HCM	Build a binary search tree for the words enology, phrenology, campanology, ornithology, ichthyology, limnology, alchemy, and astrology using alphabetical order. Using pre-order traversal to give the order of the words afterward.
150	54	CQ1_DN	How to code a text using a prefix tree? What is the algorithm to convert from an expression from infix form to postfix one?
151	54	CT_CQ1	Give as many examples as possible for representing some expressions of your choice in prefix and postfix notations.
152	54	CQ1_HCM	Give two different infix expressions that have the same calculating result after applying in-order traversal to the binary expression trees.
153	54	CQ2_HCM	What are main benefits of prefix and postfix forms?

No.	Session No	Name	Details
154	55	CQ1_DN	How many ways to build a spanning tree for a connected simple graph? Can we use greedy approachs to design algorithms for finding minimum spanning trees?
155	55	CT_CQ1	What is a spanning tree of a simple graph? Which simple graphs have spanning trees? Describe at least two different applications that require that a spanning tree of a simple graph be found.
156	55	CQ1_HCM	What are spanning trees?
157	55	CQ2_HCM	How many edges must be removed from a connected graph with n vertices and m edges to produce a spanning tree?
158	56	CQ1_DN	How many ways to build a spanning tree for a connected simple graph? Can we use greedy approachs to design algorithms for finding minimum spanning trees?
159	56	CT_CQ1	Describe depth-first search and breadth-first search algorithms for finding a spanning tree in a simple graph, and illustrate how they work for some graphs of your choice with at least eight vertices and 15 edges.
160	56	CQ1_HCM	Consider the DFS (Depth First Search) and BFS (Breadth First Search) algorithms to construct a spanning tree with the same root for a connected graph. Which algorithm will produce a higher tree?
161	57	CQ1_DN	How many ways to build a spanning tree for a connected simple graph? Can we use greedy approachs to design algorithms for finding minimum spanning trees?
162	57	CT_CQ1	What is a minimum spanning tree of a connected weighted graph? Describe at least two different applications that require that a minimum spanning tree of a connected weighted graph be found.
163	57	CQ1_HCM	What are minimum spanning trees?
164	57	CQ2_HCM	Give real-life examples of the problem of finding the minimum spanning tree.
165	58	CT_CQ1	Describe Kruskal’s algorithm and Prim’s algorithm for finding minimum spanning trees and illustrate how Kruskal’s algorithm and Prim’s algorithm are used to find a minimum spanning tree, using a weighted graph with at least eight vertices and 15 edges.
166	58	CQ1_HCM	Illustrate step by step how Kruskal’s algorithm and Prim’s algorithm are used to find a minimum spanning tree, using a weighted graph with at least 6 vertices and 9 edges.

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide
1	Progress Test	quiz	3	30.0%	>0	30 minutes	PT 1: LO1, LO2, LO6 PT 2: LO3, LO4, LO5, LO6 PT 3: LO7, LO8	OP1: MC or a suitable format // OP2: (For Constructivism Approach only): Follow lecturer's proposal	15 questions within 30 minutes	PT 1: Chapter 1, Chapter 2, Chapter 5 PT 2: Chapter 3, Chapter 4, Chapter 7 PT 3: Chapter 9, Chapter 10	OP1: by instructor, and by suitable means (computer, paper, oral...) OP2: (For Constructivism Approach only): Follow lecturer's proposal
2	Assignments/Exercises	on-going	2	20.0%	>0	at home/in class	Assignment 1: LO1, LO2, LO6 Assignment 2: LO3, LO4, LO5, LO6	OP1: Open // OP2: (For Constructivism Approach only): Follow lecturer's proposal	At least 5/ chapter	Assignment 1: Chapter 1, Chapter 2, Chapter 5 Assignment 2: Chapter 3, Chapter 4, Chapter 7	OP1: Each assignment is divided into two parts: the take-home component counts for no more than 50% of the total grade, and the in-class component counts for at least 50% (this may take the form of a presentation, essay, problem-solving, etc.). OP2: (For Constructivism Approach only): Follow lecturer's proposal

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide
3	Programming Assignment	on-going	1	10.0%	>0	at home/in class	LO3, LO4, LO7, LO8	OP1: Open // OP2: (For Constructivism Approach only): Follow lecturer's proposal	At least 1/ chapter	Chapter 3, Chapter 4, Chapter 9, Chapter 10	Students are expected to write codes to implement algorithms including, but not restricted to, those covered in the scope. Instructor may combine with other in-class assessment methods to grade the computer assignment, in which the work (source files,...) should accounts for up to 50% of the total grade.
4	Final Exam	Final exam	1	40.0%	4	60'		Computer gradable	50	All studied chapters. At least 6 questions/chapter	by Exam Board